Yaadhav Raaj

Robotics • AI • Perception • Engineer

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Senior Robotics Engineer with a decade of experience architecting, leading, and scaling real-time robotics, perception and physical AI systems. Proven track record of guiding cross-functional teams to bring advanced algorithms out of simulation and into highly constrained production hardware, with patents and publications in top journals. Deep expertise spanning the autonomy stack, from C++ middleware and OS-level integration to hardware accelerators and open-source platform contributions.

Core Competencies & Skills

Languages & Core: Modern C++ (C++20/23), Python, CUDA, OpenCL, OpenGL, Multi-threading, IPC, Real-Time Systems.
Robotics & Middleware: ROS / ROS 2, GRPC, Pinocchio, Mujoco, Eigen, GTSAM, MoveIt, Rigid Body Kinematics, Lie Theory, Sensor Fusion, Calibration
AI & Computer Vision: PyTorch, OpenCV, Object Detection, 6DoF Pose Estimation, Factor Graphs, Vision-Language-Action Models, System1/2 Architectures
Platform & Tooling: Linux, Embedded GPUs (NVIDIA/Jetson), Docker, Kubernetes, AWS, Mujoco, Unreal Engine, Git, Pipelines.
Leadership: Cross-stack system architecture, technical mentorship, design reviews, open-source ecosystem contribution.

Experience

Amazon Robotics – (Robotics Engineer - Seattle, WA) *Jan 25 – Current*

- Led the architectural design and scaling of core robot motion capabilities in inverse kinematics and motion planning
- Drove cross-stack technical execution through complex design reviews, ensuring safe, maintainable, and robust robot behavior at scale
- Building generative AI applications for automated test generation to improve simulation workflows, logging, and on-target validation
- Involved in the development and deployment of the Bluejay program

Agility Robotics – (Perception Engineer - Portland, OR) *Jun 21 – Dec 24*

- Led the perception engineering team to architect and deploy the object detection and tracking framework on the Digit humanoid robot platform.
- Directed the hardware-software integration of embedded NVIDIA GPUs, optimizing deep learning models (ONNX/TensorRT) and custom Libtorch CUDA kernels to run efficiently on highly constrained edge accelerators.
- Designed fault-tolerant systems handling sensor degradation and changing operating conditions; our perception workflows are actively scaling in production at Amazon and GXO, moving over 1,000 totes daily.
- Developed patented algorithms linking robot kinematics and environment constraints for optimal 6DoF multi-object tracking, requiring tight synchronization between hardware actuators, sensors, and middleware.
- Spearheaded the end-to-end tooling for data collection, annotation partnerships, and synthetic data generation, scaling to over 2 million unique segmentation instances in 6-DOF.

CMU Robotics Institute – (Graduate Student + Research Engineer - Pittsburgh, PA) *Feb 18 – May 21*

Under: Dr. Yaser Sheikh (CMU / Facebook Reality Labs) and Dr. Srinivas Narashiman (CMU)

- Open-Source Leadership: Core Developer for OpenPose (30k+ GitHub stars), serving as the primary library maintainer for thousands of developers globally across Linux, OSX, and Windows.
- Contributed upstream improvements and reference implementations, rewriting differentiable ADAM / SMPL models into PyTorch/CUDA and developing custom OpenCL kernels for cross-platform performance optimization.
- Hand-built and calibrated a custom Lidar, Stereo, and Programmable Light Curtain rig on a test vehicle, establishing low-level hardware synchronization and data collection pipelines.
- Published top-performing Spatio-Temporal Affinity Fields tracking system (Oral, CVPR 2019) and developed foundational probabilistic depth field estimation algorithms (CVPR 2021).

Uber ATG – (Software Engineering Intern - Pittsburgh, PA) *Jun 20 – Aug 20*

- Built upon the MultiXNet Architecture to make a Path Relative Metric and equivalent Spatio-Temporal Path Relative Field developed by my team, end-end differentiable, with custom Pytorch + CUDA modules via five point stencil

SpeedCargo – (Robotics Engineer - Singapore) *Jan 16 – Jan 18*

- Led the vision and perception development for an automated aviation cargo palletization system leveraging Gudel and ABB industrial gantry robots.
- Architected a sub-centimeter precision object measurement and mesh reconstruction system over massive workspaces, orchestrating hardware calibration across TOF and RGB sensors.
- Deployed the production-ready solution to the Singapore Changi Airport, successfully dimensioning thousands of highly variable cargo items daily using custom OpenCL/CUDA Octree implementations and particle filter optimizations.

Bumblebee Robotics - (Software Developer - Singapore) *Jun 14 – Jun 16*

- Developed patented calibration and localization algorithms fusing Sonar, Camera, IMU, and DVL sensors for the BBAUV and BBASV underwater platforms.

Education

Carnegie Mellon University – Pittsburgh, PA | Master of Science in Robotics (Aug 2019 – May 2021)
National University of Singapore – Singapore | Bachelor of Engineering in Computer Engineering (Aug 2011 – Feb 2016)

Select Patents and Publications

Select Patents: Generating object references from perception data in unstructured environments (US12552027B1) | Generating references for robot-carried objects (US12454065B1) | Method/System for AUV object localization (WO2021045679A1).

Select Papers: Exploiting and Refining Depth Distributions with Triangulation Light Curtains (CVPR 2021) | Active Perception using Light Curtains (ECCV 2020) | Single-network whole-body pose estimation (ICCV 2019) | Efficient Online Multi-Person 2D Pose Tracking (CVPR 2019)